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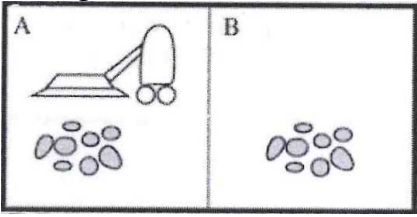
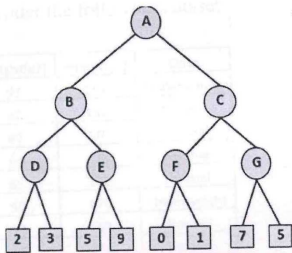
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RV COLLEGE OF ENGINEERING®
(An Autonomous Institution affiliated to VTU)
VI Semester B. E. Grade Improvement Examinations November 2021
Computer Science and Engineering
Artificial Intelligence and Machine Learning

Time: 03 Hours**Maximum Marks: 100****Instructions to candidates:**

Answer any FIVE full questions out of TEN. Each carries 20 marks.

1	1.1	What does an agent's choice of action at any given instant depends on?	01
	1.2	What are the factors that are considered for judging the Search algorithms?	01
	1.3	Give the PEAS description for a garbage picking robot.	02
	1.4	An interference algorithm that derives only entailed sentences is called _____.	01
	1.5	In the real world, most of the constraint satisfaction problems have _____ domains.	01
	1.6	Consider the following notations: G(x): x is a gold ornament S(x): x is a silver ornament P(x): x is precious Use first order logic to represent the statement "Gold and silver ornaments are precious"	02
	1.7	Categories of _____ are often characterized by structural relations among parts.	01
	1.8	What is a deterministic node?	01
	1.9	What is the input and output of Gibbs sampling algorithm?	02
	1.10	What value is used to adjust the weights a small amount in the direction that reduces the error on the training examples?	01
	1.11	The _____ characterizes the (im)purity of an arbitrary collection of examples.	01
	1.12	Consider a robot driving learning problem. What is the Performance measure and Training experience?	02
	1.13	Bring out the significance of the word "local" in locally weighted regression.	01
	1.14	Give one disadvantage of instance-based approaches.	01
	1.15	In reinforcement learning, under what condition does the agent prefer state s1 over state s2? Give reason for the preference.	02
2	a	How can a task environment be described? Discuss with examples different classifications of environments.	10
	b	Write pseudocode for an agent program. With the help of a block diagram, explain the working of a learning agent.	10
3	a	Define Artificial Intelligence from the perspective of (i) Acting Humanly and (ii) Acting Rationally.	

		Consider a vacuum-cleaner world with just two locations as given in the Fig. 3.a. Construct a partial tabulation of a simple agent function for the vacuum-cleaner world and explain the scenario.  Fig. 3.a	10
	b	Use pseudocode and block diagram to explain simple reflex agents and model based reflex agents.	10
4	a	How can a game be defined as a search problem? Define a game tree and draw a sample (partial) game tree for Tic Tac Toe.	10
	b	What is Minimax algorithm? Comment on its properties with respect to completeness, optimality, time complexity and space complexity. Write an algorithm for calculating minimax decisions.	10
5	a	Represent the following sentences in English in First Order Logic i) Not all students like both Mathematics and Science ii) Every student who loves learning is happy iii) Someone has visited every country in the world except Ethiopia iv) No one has climbed every mountain in the Himalayas	10
	b	Perform alpha Beta Pruning on the given tree in Fig. 5.b and show the steps involved. Show the ordering in which the branches might be pruned.  Fig. 5.b	10
6	a	What is Event Calculus? Give the complete set of predicates of the event calculus. Using relevant examples, demonstrate the usage of the functions that are used in representing time intervals.	10
	b	Give the full specification of a Bayesian Network and draw a sample network. Give the methodology for constructing a Bayesian Network.	10
7	a	Draw a sample Sematic Network with any 4 objects, 4 categories and relevant relationships between them.	06
	b	What are description logics? Give the syntax for CLASSIC descriptions.	06
	c	Explain how Bayesian Networks deal continuous variables? What are the mathematical concepts used in this regard?	08
8	a	With the help of a diagram, bring out the steps involved in designing a learning system.	12
	b	For the below given data set, use decision tree algorithm to find the attribute to be chosen as root node for the tree	

		<table><tr><th>Patient ID#</th><th>Sore Throat</th><th>Fever</th><th>Swollen Glands</th><th>Congestion</th><th>Headache</th><th>Diagnosis</th></tr><tr><td>1</td><td>Yes</td><td>Yes</td><td>Yes</td><td>Yes</td><td>Yes</td><td>Strep throat</td></tr><tr><td>2</td><td>No</td><td>No</td><td>No</td><td>Yes</td><td>Yes</td><td>Allergy</td></tr><tr><td>3</td><td>Yes</td><td>Yes</td><td>No</td><td>Yes</td><td>No</td><td>Cold</td></tr><tr><td>4</td><td>Yes</td><td>No</td><td>Yes</td><td>No</td><td>No</td><td>Strep throat</td></tr><tr><td>5</td><td>No</td><td>Yes</td><td>No</td><td>Yes</td><td>No</td><td>Cold</td></tr><tr><td>6</td><td>No</td><td>No</td><td>No</td><td>Yes</td><td>No</td><td>Allergy</td></tr><tr><td>7</td><td>No</td><td>No</td><td>Yes</td><td>No</td><td>No</td><td>Strep throat</td></tr><tr><td>8</td><td>Yes</td><td>No</td><td>No</td><td>Yes</td><td>Yes</td><td>Allergy</td></tr><tr><td>9</td><td>No</td><td>Yes</td><td>No</td><td>Yes</td><td>Yes</td><td>Cold</td></tr><tr><td>10</td><td>Yes</td><td>Yes</td><td>No</td><td>Yes</td><td>Yes</td><td>Cold</td></tr></table>	Patient ID#	Sore Throat	Fever	Swollen Glands	Congestion	Headache	Diagnosis	1	Yes	Yes	Yes	Yes	Yes	Strep throat	2	No	No	No	Yes	Yes	Allergy	3	Yes	Yes	No	Yes	No	Cold	4	Yes	No	Yes	No	No	Strep throat	5	No	Yes	No	Yes	No	Cold	6	No	No	No	Yes	No	Allergy	7	No	No	Yes	No	No	Strep throat	8	Yes	No	No	Yes	Yes	Allergy	9	No	Yes	No	Yes	Yes	Cold	10	Yes	Yes	No	Yes	Yes	Cold	08													
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9	a	When do we consider that computer program learns? Identify P,T and E for a chess learning program, an autonomous driving robot and a hand writing learning program?	10																																																																																										
b	Consider the below given dataset D, compute the following: Class-Labeled Training Tuples from the <i>AlIElectronics</i> Customer Database																																																																																												
		<table><tr><th>RID</th><th>age</th><th>income</th><th>student</th><th>credit_rating</th><th>Class: buys_computer</th></tr><tr><td>1</td><td>youth</td><td>high</td><td>no</td><td>fair</td><td>no</td></tr><tr><td>2</td><td>youth</td><td>high</td><td>no</td><td>excellent</td><td>no</td></tr><tr><td>3</td><td>middle_aged</td><td>high</td><td>no</td><td>fair</td><td>yes</td></tr><tr><td>4</td><td>senior</td><td>medium</td><td>no</td><td>fair</td><td>yes</td></tr><tr><td>5</td><td>senior</td><td>low</td><td>yes</td><td>fair</td><td>yes</td></tr><tr><td>6</td><td>senior</td><td>low</td><td>yes</td><td>excellent</td><td>no</td></tr><tr><td>7</td><td>middle_aged</td><td>low</td><td>yes</td><td>excellent</td><td>yes</td></tr><tr><td>8</td><td>youth</td><td>medium</td><td>no</td><td>fair</td><td>no</td></tr><tr><td>9</td><td>youth</td><td>low</td><td>yes</td><td>fair</td><td>yes</td></tr><tr><td>10</td><td>senior</td><td>medium</td><td>yes</td><td>fair</td><td>yes</td></tr><tr><td>11</td><td>youth</td><td>medium</td><td>yes</td><td>excellent</td><td>yes</td></tr><tr><td>12</td><td>middle_aged</td><td>medium</td><td>no</td><td>excellent</td><td>yes</td></tr><tr><td>13</td><td>middle_aged</td><td>high</td><td>yes</td><td>fair</td><td>yes</td></tr><tr><td>14</td><td>senior</td><td>medium</td><td>no</td><td>excellent</td><td>no</td></tr></table> i) The expected information gain needed to classify a tuple in D ii) Gain in the information by partitioning the tree with ‘Age’ as the node attribute and with ‘Income’ as the node attribute.	RID	age	income	student	credit_rating	Class: buys_computer	1	youth	high	no	fair	no	2	youth	high	no	excellent	no	3	middle_aged	high	no	fair	yes	4	senior	medium	no	fair	yes	5	senior	low	yes	fair	yes	6	senior	low	yes	excellent	no	7	middle_aged	low	yes	excellent	yes	8	youth	medium	no	fair	no	9	youth	low	yes	fair	yes	10	senior	medium	yes	fair	yes	11	youth	medium	yes	excellent	yes	12	middle_aged	medium	no	excellent	yes	13	middle_aged	high	yes	fair	yes	14	senior	medium	no	excellent	no	10
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10	a	Give the K-NN algorithm for training and classification. Consider the following dataset	10																																																																																										
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	b	How does reinforcement learning problem differs from other function approximation problems? With the help of mathematical equations and pseudocode, explain Q-learning.	10																																																																																										